

Research Paper

Experimental study of a chemisorption heat pump under different operation conditions

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ARTICLE INFO

Keywords:

Chemisorption
Heat pump
Cooling capacity
Coefficient of performance
Specific cooling power
Optimal operation condition

ABSTRACT

A chemisorption heat pump utilizes thermal energy for cooling. Chemisorption heat pump can be applied to a wider variety of applications when the system uses thermal energy at lower temperature. Thus, study on the performance of a chemisorption heat pump operated by thermal energy of 40 °C was conducted. 500 W system was built and its performance was analyzed through experiments. Chemisorption heat pumps are capable of continuous cooling operation through alternating operation, in which the system performance changes depending on the operating time of each operating mode. There are no existing studies have analyzed the performance of a chemisorption heat pump driven by a 40 °C heat source according to the operating time. Therefore, the performance change according to the operation time of each modes was identified. The cooling capacity, coefficient of performance and specific cooling power were measured in corresponding ranges of 272.2–543.2 W, 0.06–0.24 and 42.5–84.9 W/kg for each case. The cooling capacity is maximum when the operation times of the 'adsorption/desorption' and 'preparation 1' mode are 10 min and 10 s, respectively, and the coefficient of performance is maximum when the operation times are 30 min and 20 s, respectively.

1. Introduction

In South Korea, thermal energy consumption is estimated to account for approximately 27 % of all energy consumption [1]. Most thermal energy is not recycled once used. In order to increase national energy efficiency, research on waste heat recovery is being conducted in various fields. According to H. Jouhara *et al.* [3], most studies related to waste heat recovery focus on processes that occur at temperatures exceeding 100 °C, and not many studies have examined waste heat recovery in the range of 40–50 °C. According to Arpagaus *et al.* [4], thermal energy in various temperature bands of 40–300 °C is required for paper, chemical, and food processes, meaning that waste heat at a temperature of at least 40 °C is generated from various industrial processes. Therefore, if thermal energy at a low temperature of 40–50 °C can be recycled, the overall energy utilization efficiency can be increased.

Power outages are constantly occurring in the world. In 2011, large-scale blackouts occurred in South Korea, and in Taiwan, large-scale blackouts occurred in 2017 and 2021. In South Korea, the amount of spare electricity decreases every summer due to increased electric loads. Air conditioning systems are one of the causes of the increased electrical load in summer.

If cooling can be accomplished by recycling waste heat in summer, energy efficiency improvements and electric load reductions can be achieved at the same time. This concept motivates research on thermal driven heat pump systems. Two representative thermal driven heat pumps are the adsorption heat pump and the absorption heat pump. For the absorption heat pump, thermal energy at a temperature close to 100 °C is required for the system to operate, and many studies have concentrated on this temperature band [5]. In case of the adsorption heat pump, it can be driven based on heat sources in various temperature bands according to the combination of the adsorbent and working fluid used. According to Yang *et al.* [6], a certain combination of an adsorbent and a working fluid can facilitate operation based on thermal energy of 40–50 °C. The adsorbent and working fluid combination for use in this temperature band relies on a chemisorption method in which the working fluid is adsorbed and desorbed into the adsorbent by chemical force.

Various studies have conducted on the chemisorption heat pump systems. Previous research can be divided into studies related to adsorbent and refrigerant, chemisorption cycle analysis, chemisorption cycle experiments. In case of the researches related to adsorbent and refrigerant, Han *et al.* [7] measured the effective thermal conductivity and gas permeability of the adsorbent used and conducted a simulation

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Received 25 September 2023; Received in revised form 12 December 2023; Accepted 19 December 2023

Available online 22 December 2023

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Nomenclature		x	Conversion ratio
Ar	Pre-exponential factor	<i>Greek letters</i>	
c_p	Specific heat	θ	Rotation angle
COP	Coefficient of performance	<i>Subscripts</i>	
dT ₀	Operation time of the 'Adsorption/desorption' mode	2nd	Secondary
dT ₁	Operation time of the 'Preparation I' mode	avg	Average
E _a	Activation energy	cycle	One cycle for each mode
P	Pressure	dr	Desorption reactor
Q	Heat transfer rate	eq	Equilibrium
R	Gas constant	eva	Evaporator
SCP	Specific cooling power	i	Inlet
T	Temperature	o	Outlet
Δt	Time interval	sampling	Data sampling
t	Time		
w	Flow rate		

of a reactor based on the measured thermophysical properties. Zhou *et al.* [8] measured the non-equilibrium line of chlorides-ammonia working pairs for a refrigeration system. According to their paper, the refrigeration process is the non-equilibrium process, meaning that a non-equilibrium line is needed for precise predictions of the performance of a chemisorption heat pump. Y. Yuan *et al.* [9] performed an adsorption/desorption experiment of ammonia and SrCl_2 and presented a kinetic model.

Wang *et al.* [10] reviewed adsorption working pairs for refrigeration systems. They considered physical and chemical adsorption working pairs and compared the coefficient of performance (COP) and specific cooling power (SCP) of each type of working pairs. L. F. Cabeza *et al.* [11] reviewed research results based on adsorbent and refrigerant, and in this paper, they summarized the thermal properties of the adsorbent and system performance according to the adsorbent /refrigerant combination.

Some studies conducted performance analysis through the modeling. Aristov [12] conducted analytical research on an adsorption heat pump, suggesting the theoretical efficiency of both chemical heat pumps and adsorption heat pumps. Oliveira [13] studied a chemisorption heat pump for water heating and steam production. He suggested the cycle and working pairs and conducted a performance analysis of the heat pump system. Yang *et al.* [14] developed dynamic model for a natural-gas-powered chemisorption heat pump system. They suggested the heating capacity and COP of the chemisorption heat pump based on simulation results. Yang *et al.* [15] conducted a performance analysis of a chemisorption heat pump with different working pairs and cycle configurations. Bao *et al.* [16] built a dynamic model of a chemisorption refrigeration and power generation system. In their paper, the temperature of the heat source had to be high to generate power. Although not focusing on heat pumps, thermal cycle analyses using the principle of chemisorption were conducted by several researchers [17,18]. Oro *et al.* [19] conducted a performance analysis of a chemisorption chiller supplied with a heat source from a fuel cell system. They developed a dynamic model of a chemisorption chiller and discussed the power and cooling capacities of the overall system. They considered NaBr and ammonia as the working pair, and the temperature of the heat source was close to 70 °C. X. Xie *et al.* [20] conducted modeling on the chemisorption-compression hybrid high temperature heat pump system and conducted performance analysis. H. Hi *et al.* [21] suggested start-stop method of chemisorption heat pump based on the simulation results.

Looking at experimental studies on the chemisorption cycle, Pachó [22] conducted an analytical and experimental study of a chemisorption heat pump that used carbon and ammonia as the adsorbent and working fluid, respectively. In this study, the temperature of the high-

temperature thermal energy was about 120 °C. Huang *et al.* [23] focused on the reactor of the chemisorption heat pump. They conducted analytical and experimental research on the chemical reaction process under isobaric conditions. Oliveira and Generoso [24] developed chemisorption heat pump system driven by hot water. They conducted an experiment and suggested the COP and cooling power of their chemisorption heat pump. The temperature of the hot water was approximately 70 °C. Kim *et al.* [25] conducted an experimental study of the performance of a chemisorption heat pump driven by a heat source at 40 °C. The cooling capacity of the system approached 100 W. P. Gao and L. Wang [26] conducted experiments on heating cycles in cold regions and analyzed system performance.

In addition, there are several papers summarizing overall research related to chemisorption system. Li *et al.* [27] reviewed the research progress on solid-gas sorption refrigeration systems. They summarized studies of various types of adsorption refrigeration cycles. Pinheiro *et al.* [28] reviewed the literature on adsorption heat pumps for heating applications. Wang *et al.* [29] wrote a textbook on chemisorption heat pump systems. They explained the properties of adsorbents, the kinetics of adsorbents, and the cycle configurations used with refrigeration systems.

As seen above, researches from various perspectives have been conducted on chemisorption heat pumps. Among these, the heat source temperatures in most previous studies focused on the chemisorption heat pump cycle was above 60 °C. Chemisorption heat pumps driven by heat sources around temperatures of 40 °C have not been studied as much. The lower the temperature of thermal energy used to drive the system, the more applications it can be applied to. There is also a lack of research analyzing system performance under different operation time. Oliveira and Generoso [24] presented performance changes over cycle time, but did not evaluate the performance change according to the operation time of each operation modes. In order to accurately characterize the operation of a chemisorption heat pump, it is necessary to quantitatively analyze the impact of the operating time of each operating mode on the system performance. This allows user to set up the operation time of the chemisorption heat pump according to their purpose.

In this paper, an experimental study was conducted to quantitatively analyze the performance change according to the alternating operation time for a chemisorption heat pump system driven by thermal energy of 40 °C and to derive the optimal operation point. A 500 W class chemisorption heat pump was built, and the cooling capacity, COP, and SCP were measured according to the operating time of 'adsorption/desorption' mode and 'preparation' mode. Based on the experimental results, operating conditions for optimal cooling capacity and that for optimal COP were presented.

2. Experimental method

2.1. Experimental apparatus

Fig. 1 shows a schematic diagram of the experimental apparatus. The system consists of two reactors, a condenser, and an evaporator. Fig. 2 shows drawings of the reactor and heat exchangers. The design values of each component are described in Table 1. Ammonia and NaBr, which are known to be suitable, respectively, as the working fluid and adsorbent of a chemisorption heat pump under a low temperature condition of 40 °C, were used. For the reactor, finned tubes are installed in the shell. The adsorbent fills the spaces between the fins and is wrapped into a mesh to prevent it from escaping. NaBr impregnated into expanded graphite was used as an adsorbent. The mass ratio of expanded graphite to NaBr was set to 1:4. Each fin tube was filled with 100 g of adsorbent, and 3.2 kg of adsorbent in total was used to fill each reactor. Fig. 3 shows an image of the experimental apparatus. All components and piping lines are wrapped in an insulating material.

The internal temperatures, internal pressures, and inlet/outlet temperatures of the reactor, condenser and evaporator were measured. The inlet/outlet temperatures, inlet pressure, and flow rate of each secondary fluid were also measured. Specifications of each sensor are described in Table 2. All sensors were calibrated by a certified authority. The GL820 model by DATAQ Instruments was used as a data logger.

Alternating operation is essential for chemisorption heat pumps. An automatic valve is required to realize this type of operation, and 13 automatic valves in total were installed in this experimental apparatus. Four chillers were used to supply secondary fluid to the reactors and heat exchangers. The specifications of each chiller are given in Table 2.

2.2. Experimental procedure and conditions

Fig. 4 shows the Clapeyron curve of the chemisorption heat pump system. In order to operate the system, it is necessary to supply high- and low-temperature secondary fluids to the desorption reactor and evaporator, respectively. In addition, secondary fluids at a medium temperature should be supplied to the condenser and to the adsorption reactor. Chiller 1 and Chiller 3 are used to supply secondary fluid at a medium temperature to the adsorption reactor and condenser, respectively. Chillers 2 and 4 supply secondary fluid at a high and low temperature to the desorption reactor and evaporator, respectively.

In this research, there are six operation modes, and the valve positions under each mode are shown in Table 3. All valves were hydraulic on/off valves and were programmed to operate automatically at the start of each mode. Each reactor alternately acts as a desorption and adsorption reactor according to each mode.

During mode 1, valve 1, 4, 6, 8, 11, 13 are opened. Reactor 1 is connected to the evaporator, and secondary fluid at a medium temperature is supplied to reactor 1. Meanwhile, reactor 2 is connected to the condenser, and secondary fluid at a high temperature is supplied to reactor 2. Reactors 1 and 2 act as an adsorption reactor and a desorption reactor, respectively. The working fluid is desorbed from the adsorbent in the desorption reactor and flows to the condenser and is condensed. Hence, the amount of working fluid (liquid phase) increases in the condenser. On the other hand, the working fluid in the liquid phase is evaporated at the evaporator and flows to the adsorption reactor, after which it is adsorbed by the adsorbent. Thus, the amount of liquid-phase working fluid decreases in the evaporator. In the condenser and the evaporator, continuous alternating operation is possible only when the amount of working fluid in a liquid state is maintained. Therefore, it is necessary to move the working fluid from the condenser to the

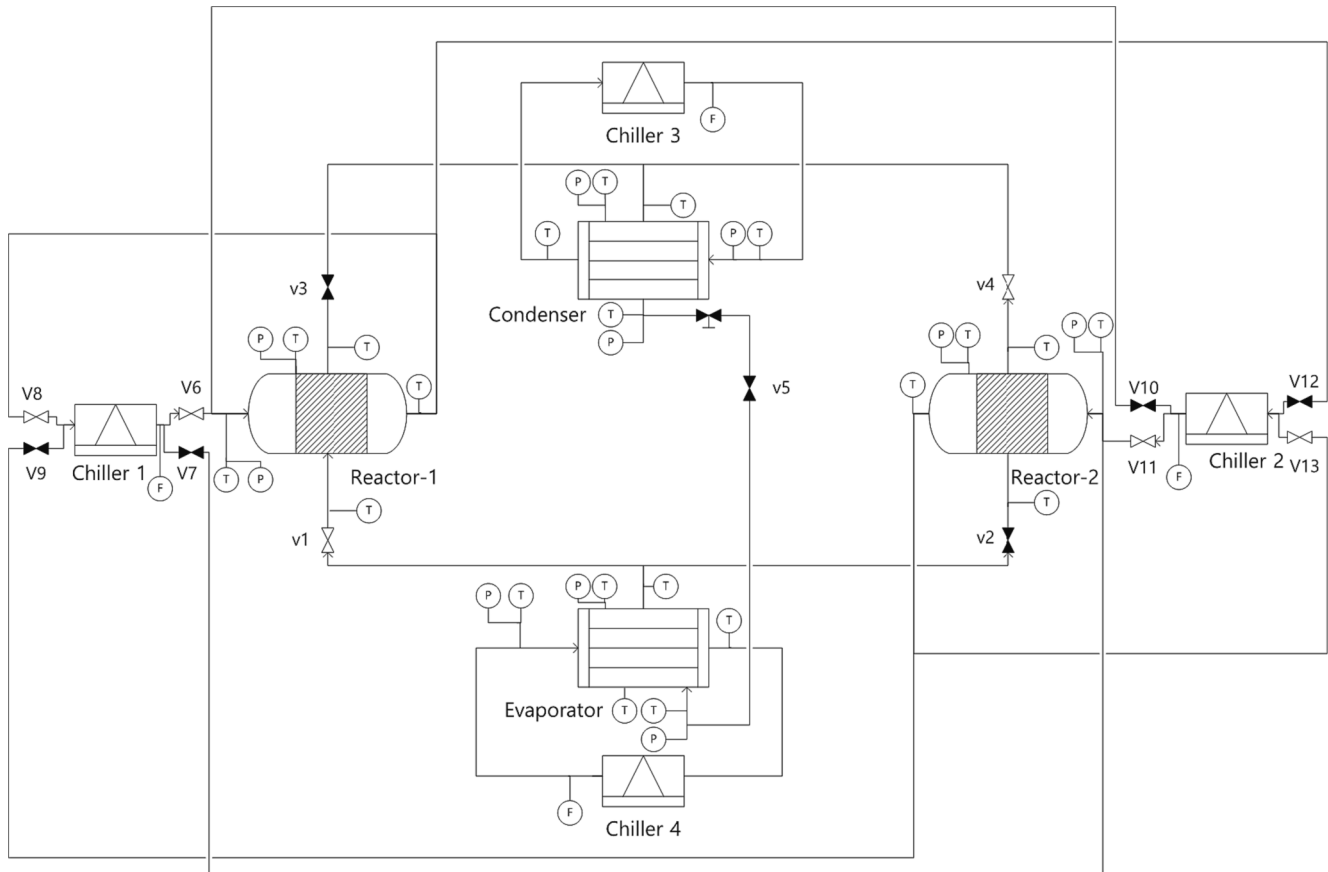


Fig. 1. Schematic diagram of the experimental apparatus.

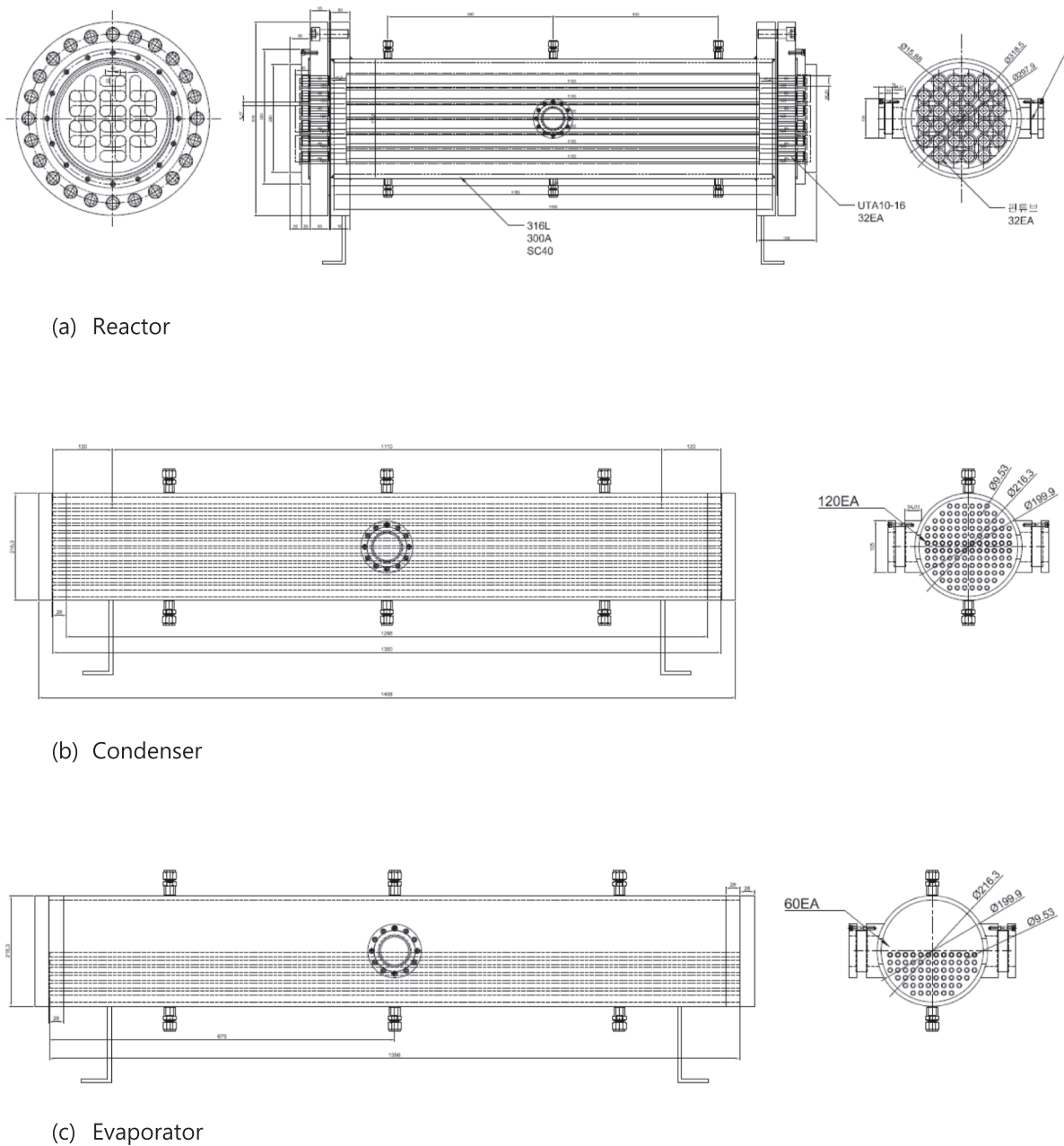


Fig. 2. Drawings of main components.

Table 1
Specifications of the Main Components.

	Reactor	Evaporator	Condenser
Shell O.D. (mm)	318.5	216.3	216.3
Shell I.D. (mm)	297.9	199.9	199.9
Tube L (mm)	1350	1350	1350
Tube O.D. (mm)	15.88	9.53	9.53
No. of tube (-)	32	60	120
Fin height (mm)	8	—	—
Fin thickness (mm)	0.8	—	—
Fin pitch (FPI)	7	—	—

evaporator.

During mode 2, valve 1, 4, 6, 8, 11, 13 are closed and valve 5, 7, 9, 10, 12 are opened. The liquid-phase working fluid moves from the condenser to the evaporator. At this time, almost all of the liquid refrigerant accumulated in the condenser moves to the evaporator. In

cases of secondary fluids supplied to reactors, the flow path of each secondary fluid to the reactors changes such that medium-temperature secondary fluid and high-temperature secondary fluid are supplied to reactor 2 and reactor 1, respectively. At the start of mode 2, the temperatures of chiller 1 and 2 fluctuate momentarily due to the heat capacity of the reactor. Since the time for the temperatures of chillers 1 and 2 to stabilize are longer than the time for the liquid refrigerant in the condenser to move to the evaporator, when the transfer of the liquid refrigerant is completed, valve 5 is closed and proceed to mode 3. In mode 3, it was waited until the temperatures of the secondary fluids supplied from each chiller stabilized.

In mode 4, valve 2 and 3 are opened and the adsorption/desorption operation is performed. Reactor 2 is connected to an evaporator and reactor 1 is connected to a condenser, so that reactor 2 becomes an adsorption reactor and reactor 1 becomes a desorption reactor. Then, Subsequently, preparation modes start again and then modes 1–6 are repeated.

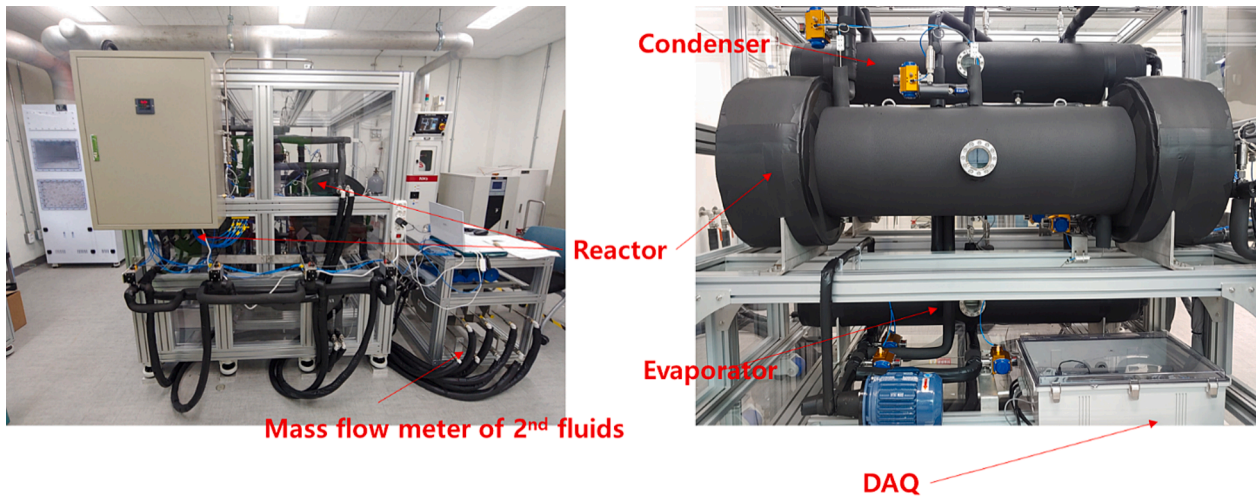


Fig. 3. Images of the experimental apparatus.

Table 2
Specifications of the Sensors and Chillers.

	Manufacturer	Model	Uncertainty
Resistance thermometer	OMEGA	PT100Ω	±0.2 °C
Pressure sensor	SENSYS	PSHB0030BCPG	±0.8 kPa
Mass flow meter	TRICOR	TCM 7900-FC-SGSS-CSMS	±0.26 % of reading
Mass flow meter	TRICOR	TCM 3100-FA-SGSS-CSMS	±0.26 % of reading
Chiller 1 and 3	HYUNDAI ENG.	HAD-010HC	–
Chiller 2 and 4	JEIOTECH	HH-55H	–

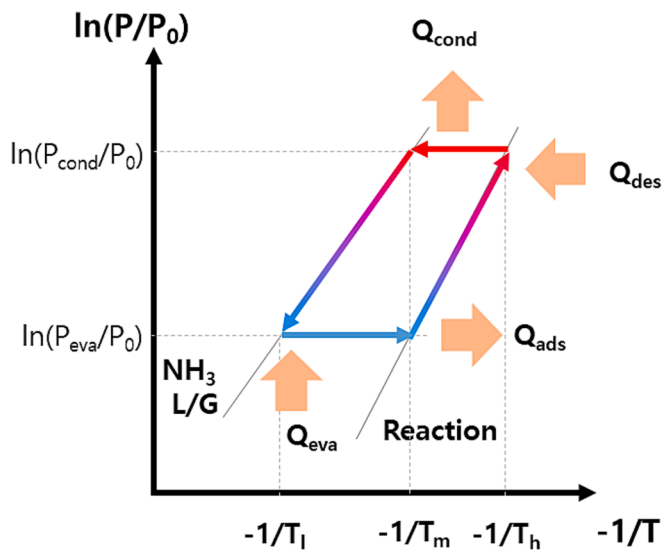


Fig. 4. Clapeyron curve of the chemisorption heat pump system [6].

The temperature and flow rate of the secondary fluid affect the system performance. The average temperatures and flow rates of the secondary fluid for the evaporator, condenser, adsorption reactor and desorption reactor are shown in Table 4. The secondary fluids supplied from chillers 1 to 4 are a mixture of water and ethylene glycol, and the concentration of ethylene glycol is 50 %.

In this research, the intent was to understand the performance according to the operation time for each mode. The experiment conducted

here tested 21 conditions in total. The operating times for each mode under each condition are shown in the Table 5. The operating times of modes 4–6 are identical to those of modes 1–3.

3. Results and discussion

3.1. Data reduction

For each experimental condition in Table 5, modes 1–6 in Table 3 were repeatedly operated three times in total. A data analysis was conducted based on the experimental results of the second and third times because the alternating operation may not be stabilized during the first repeated operation results.

Fig. 5 shows the experimental results of case 15 in Table 5. In this research, 'Preparation I' to 'Adsorption/desorption' in Table 3 are defined as one cycle. In Fig. 5, the black dotted line represents each cycle boundary. The first and third cycles in Fig. 5 are the experimental results of modes 5, 6 and 1 in Table 3. The second and fourth cycles of Fig. 5 are the experimental results of modes 2, 3 and 4 in Table 3. As indicated by the experiment results, the temperature and pressure results change repeatedly and constantly during each cycle. In order to evaluate the performance of the system in a chemisorption heat pump, the heat transfer rate of the evaporator and the desorption reactor should be identified. In this study, the heat transfer rates of the evaporator and desorption reactor were calculated based on the temperatures and flow rates of the secondary fluids supplied to the evaporator and desorption reactor. The instantaneous heat transfer rate at a specific point in time was calculated based on Eqs. (1) and (2) below.

$$Q_{eva} = (wc_p)_{eva,2nd} (T_{eva,2nd,i} - T_{eva,2nd,o}) \quad (1)$$

$$Q_{dr} = (wc_p)_{dr,2nd} (T_{dr,2nd,i} - T_{dr,2nd,o}) \quad (2)$$

In the chemisorption heat pump, the adsorption/desorption rate varies according to the adsorption amount of working fluid in the adsorbent during the adsorption/desorption operation process. Therefore, the instantaneous heat transfer rate of the chemisorption heat pump changes continuously during the adsorption/desorption process, as shown in Fig. 5. For this reason, the performance of a chemisorption heat pump should be evaluated using the average heat transfer rate. The average heat transfer rate during each cycle was evaluated based on Eqs. (3) – (4).

$$Q_{eva,avg} = \frac{\sum_{i=1}^n Q_{eva,i} \Delta t_{sampling}}{t_{cycle}} \quad (3)$$

Table 3

Valve Positions under Each Operation Mode.

No.	Mode	Valve position												
		1	2	3	4	5	6	7	8	9	10	11	12	13
1	Adsorption (Rx1)/desorption (Rx2)	O	X	X	O	X	O	X	O	X	X	O	X	O
2	Preparation I	X	X	X	X	O	X	O	X	O	O	X	O	X
3	Preparation II	X	X	X	X	X	X	O	X	O	O	X	O	X
4	Adsorption (Rx2)/ desorption (Rx1)	X	O	O	X	X	X	O	X	O	O	X	O	X
5	Preparation I	X	X	X	X	O	O	X	O	X	X	O	X	O
6	Preparation II	X	X	X	X	X	O	X	O	X	X	O	X	O

Table 4

Flow Rates and Temperatures of the Secondary Fluid of Each Main Component.

Component	Flow rate (kg/min)	Component	Inlet temperature of secondary fluid (°C)
Reactor 1 & 2	14.4	Adsorption/ Desorption reactor	19.7/40.2
Condenser	2.7	Condenser	20.5
Evaporator	2.8	Evaporator	15.9

Table 5

Operation Times of Each Mode under All Cases.

Case No.	Operation time of Mode 1/ 2/3 (s)	Case No.	Operation time of Mode 1/ 2/3 (s)
1	300/10/290	12	900/180/120
2	300/20/280	13	900/300/0
3	300/60/240	14	1200/20/280
4	300/300/0	15	1200/60/240
5	600/10/290	16	1200/180/120
6	600/20/280	17	1200/300/0
7	600/60/240	18	1800/20/280
8	600/180/120	19	1800/40/260
9	600/300/0	20	1800/180/120
10	900/10/290	21	1800/300/0
11	900/20/280		

$$Q_{dr,avg} = \frac{\sum_{i=1}^n Q_{dr,i} \Delta t_{sampling}}{t_{cycle}} \quad (4)$$

$\Delta t_{sampling}$ was one second and t_{cycle} for each case is the summation of the operation times of modes 1–3 in Table 5. With regard to Fig. 5, t_{cycle} is 1500 s.

There are two parameters used to analyze the performance of a chemisorption heat pump: the COP and the SCP. The COP and SCP are correspondingly defined as Eqs. (5) and (6).

$$COP_{avg} = \frac{Q_{eva,avg}}{Q_{dr,avg}} \quad (5)$$

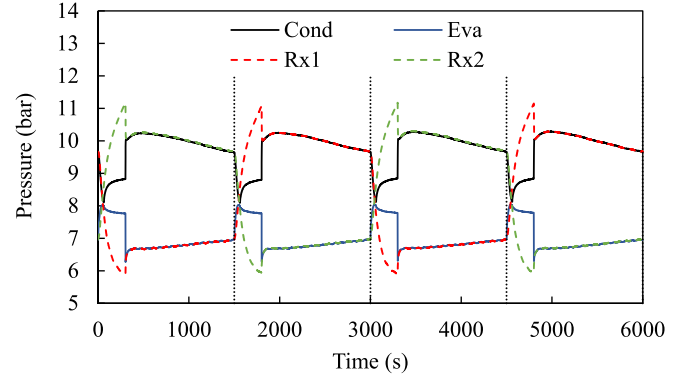
$$SCP_{avg} = \frac{Q_{eva,avg}}{m_{adsorbent} \Delta T_{wholesystem}} \quad (6)$$

As noted earlier, the mass of the adsorbent in the entire system is 6.4 kg.

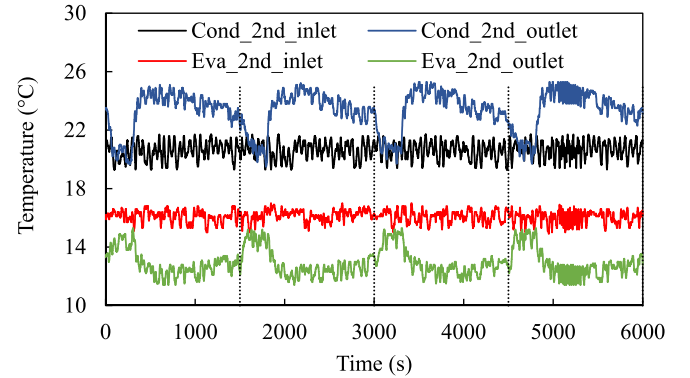
The uncertainties of the resistance temperature detector (RTD) and mass flow meter are shown in Table 2. Based on uncertainties of each sensor, the uncertainty in the values of Q_{eva} , Q_{dr} , COP and SCP were calculated as recommended by Taylor [30]. Eq. (7) is used to calculate the uncertainties of Q_{eva} , Q_{dr} , COP and SCP.

$$\Delta P = \sqrt{\left(\frac{dP}{dV_1} \Delta V_1\right)^2 + \left(\frac{dP}{dV_2} \Delta V_2\right)^2 + \left(\frac{dP}{dV_3} \Delta V_3\right)^2 + \dots + \left(\frac{dP}{dV_n} \Delta V_n\right)^2} \quad (7)$$

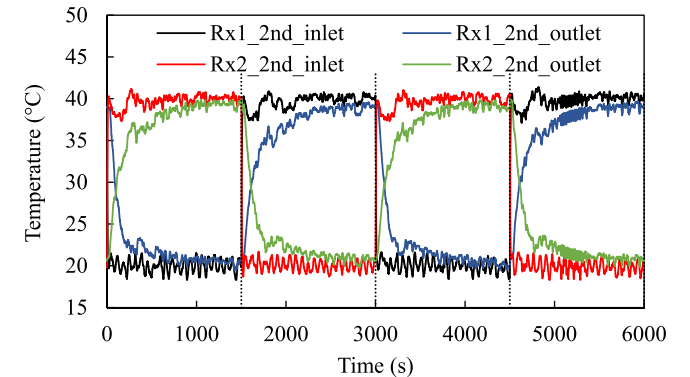
Q_{eva} , Q_{dr} , COP and SCP were obtained from Eqs. (1)–(4), respectively. In Eqs. (1) and (2), the flow rates and the inlet and outlet temperatures of



(a) Pressure of each main component



(b) Inlet and outlet temperatures of the secondary fluids of the evaporator and condenser



(c) Inlet and outlet temperatures of the secondary fluids of the adsorption and desorption reactors

Fig. 5. Experimental result of case 15.

the secondary fluids were used. Thus, the uncertainties of these sensors were used to calculate the uncertainties of Q_{eva} , Q_{dr} , COP and SCP.

3.2. Alternating operation characteristics of a chemisorption heat pump

Experiments were conducted on a total of 21 cases, and the tendencies in these experimental results are similar to those shown in Fig. 5. Fig. 5-(a) shows the internal pressures of the reactors, evaporator and condenser and Fig. 5-(b) shows the inlet and outlet temperatures of the secondary fluids of the condenser and evaporator. Fig. 5-(c) shows the inlet and outlet temperatures of the secondary fluids of each reactor.

As noted above, each cycle is from 'Preparation I' to 'Adsorption/desorption'. When the 'Preparation I' mode starts, valves 1–4 are closed and valve 5 is open, and the temperatures of the secondary fluids for each reactor change momentarily. In cycles 1 and 3, reactor 1 acts as an adsorption reactor, and reactor 2 acts as a desorption reactor. During cycle 2 and 4, reactor 1 acts as a desorption reactor and reactor 2 acts as an adsorption reactor. Reactor 1 adsorbs working fluid under a medium temperature condition during cycle 1. Before the end of cycle 1, temperature of reactor 1 is close to a medium temperature and the adsorbent in reactor 1 adsorbs large amounts of the working fluid. On the other hand, reactor 2 desorbs working fluid under the high temperature condition during cycle 1. Before the end of cycle 1, for reactor 2, the temperature of the structure is close to the high temperature and only a small amount of working fluid adsorbed in the adsorbent of reactor 2. The pressures of the evaporator and condenser are around the saturation pressure of the low and medium temperatures at the end of cycle 1, respectively. Then, cycle 2 ('Preparation I' Mode) starts. Reactors 1 and 2 are disconnected from the evaporator and condenser, respectively, and they become isolated during 'Preparation I' and 'Preparation II'. For reactor 1, the secondary fluid at a high temperature is supplied, and the temperature of the structure of reactor 1 increases. The secondary fluid at a high temperature also supplies thermal energy to the adsorbent in reactor 1. The working fluid in the adsorbent is desorbed, causing the pressure of reactor 1 to increase. Thus, the pressure of reactor 1 increases during the 'Preparation I' and 'Preparation II' modes of cycles 2 and 4. On the other hand, secondary fluid at a medium temperature is supplied to reactor 2, and temperature of the structure of reactor 2 decreases. Thermal energy is transferred from the adsorbent in reactor 2 to the secondary fluid, and the gas-phase working fluid is adsorbed by the adsorbent in reactor 2. Therefore, the pressure of reactor 2 decreases in the 'Preparation I' and 'Preparation II' modes of cycles 2 and 4. The condenser and evaporator are connected to each other during 'Preparation I'. Thus, the pressure of the condenser decreases and that of the evaporator increases in the 'Preparation I' mode. The liquid-phase working fluid flows from the condenser to the evaporator in this mode. If the 'Preparation I' mode is maintained after all of the liquid moves to the evaporator, the pressures of the evaporator and condenser are equalized. After the 'Preparation I' mode, the condenser and evaporator are disconnected from each other and the pressure of the condenser increases while that of the evaporator decreases. This tendency is shown in Fig. 5-(a). When the 'Preparation I' mode starts, the outlet temperature of the secondary fluid in the condenser decreases rapidly and the outlet temperature of the secondary fluid in the evaporator increases rapidly. Subsequently, the outlet temperatures of the secondary fluids in evaporator and condenser remain constant. In Fig. 5-(a), the pressure of the evaporator increases and that of the condenser decreases at the beginning of the 'Preparation I' mode. Thus, the internal temperatures of the condenser and evaporator also change rapidly such that the heat transfer rate also varies sharply. As noted earlier, the pressure of the adsorption reactor decreases in the 'Preparation I/II' modes and the pressure of the adsorption reactor is lower than that of the evaporator at the end of the 'Preparation II' mode. The outlet temperature of the secondary fluid in the adsorption reactor decreases sharply when the 'Preparation I' mode starts. In the adsorption reactor, the inlet temperature of the secondary fluid changes instantaneously

upon a change in the flow direction through the valve, but the outlet temperature decreases gradually due to the heat capacity of the adsorption reactor and the secondary fluid. The outlet temperature of the secondary fluid in the desorption reactor changes inversely to that of the adsorption reactor under the same principle. The heat transfer rate of the desorption reactor is calculated by Eq. (2), and the value is very high at the beginning of the 'Preparation I' mode. This is an inevitable heat loss that occurs during the alternating operation of the chemisorption heat pump.

Subsequently, the 'Adsorption/desorption' mode starts. The adsorption reactor (reactor 2 of cycles 2 and 4) and the desorption reactor (reactor 1 of cycles 2 and 4) are connected to the evaporator and condenser, respectively. The working fluid in the evaporator is vaporized and flows to the adsorption reactor. Thus, the pressure of the evaporator should be higher than that of the adsorption reactor. The outer diameter of the pipe between the evaporator and reactor is $1/2''$, which is large enough compared to the flow rate of the working fluid. Therefore, the pressure drop in the pipeline between the evaporator and adsorption reactor is very small such that the pressure of the evaporator is very slightly higher than that of the adsorption reactor. The working fluid is desorbed from the adsorbent in the desorption reactor and flows to the condenser. Thus, the pressure of the desorption reactor should be higher than that of the condenser. The outer diameter of the pipe between the condenser and the reactor is $1/2''$, and this is also large enough compared to the flow rate of the working fluid. Thus, the pressure of the desorption reactor is very slightly higher than that of the condenser. Fig. 5-(a) shows these features. The gas-phase refrigerant moves from the evaporator to the adsorption reactor and the pressures of each component become nearly identical. During this process, the temperature of the working fluid inside the evaporator also decreases sharply, which instantaneously increases the heat transfer rate in the evaporator. Thus, the outlet temperature of the secondary fluid in the evaporator decreases at the beginning of the 'Adsorption/desorption' mode. The outlet temperature of the secondary fluid in the condenser also increases rapidly by means of a similar principle. The outlet temperature of the secondary fluid in the desorption reactor increases until the end of the 'Adsorption/desorption' mode. The outlet temperature of the secondary fluid in the desorption reactor is always lower than the inlet temperature because thermal energy is supplied from the secondary fluid of the desorption reactor to the adsorbent when the working fluid is desorbed from the adsorbent. On the other hand, the outlet temperature of the secondary fluid in the adsorption reactor is always higher than the inlet temperature because thermal energy is released from the adsorbent when the working fluid is adsorbed by the adsorbent.

When conducting the experiment according to the procedure presented in Chapter 2, it can be confirmed that the pressures inside the system and the temperatures of the secondary fluids keep stable from the third cycle, as shown in the results in Fig. 5.

The average heat transfer rates of the evaporator, that of the desorption reactor, and the average COP and SCP values were calculated using Eqs. (1)–(6), and Fig. 6 shows these results. There are five legends in Fig. 6, where 'average' refers to the average value of the entire operation (from cycle 1 to cycle 4). As shown in Fig. 6-(a), the average heat transfer rate of the evaporator is approximately 513 W. The maximum difference between the average value and each cycle is close to 3.4 %. The average heat transfer rate of the desorption reactor is about 2482 W. The maximum difference between the average value and the values in each cycle is about 5.1 %. The average COP and SCP are about 0.21 and 80.1 W/kg, respectively. The maximum differences of COP and SCP between the average value and the values in each cycle are 7.7 % and 3.4 %, respectively. As indicated by the results, the performance of each cycle is maintained within a small amount of deviation. The deviation of each cycle occurs due to fluctuations of the boundary conditions and uncertainties of the sensors. For example, the flow rate of the secondary fluid in the reactor varied slightly during the cycle, as shown in Fig. 7. Although the structure of each reactor is designed and

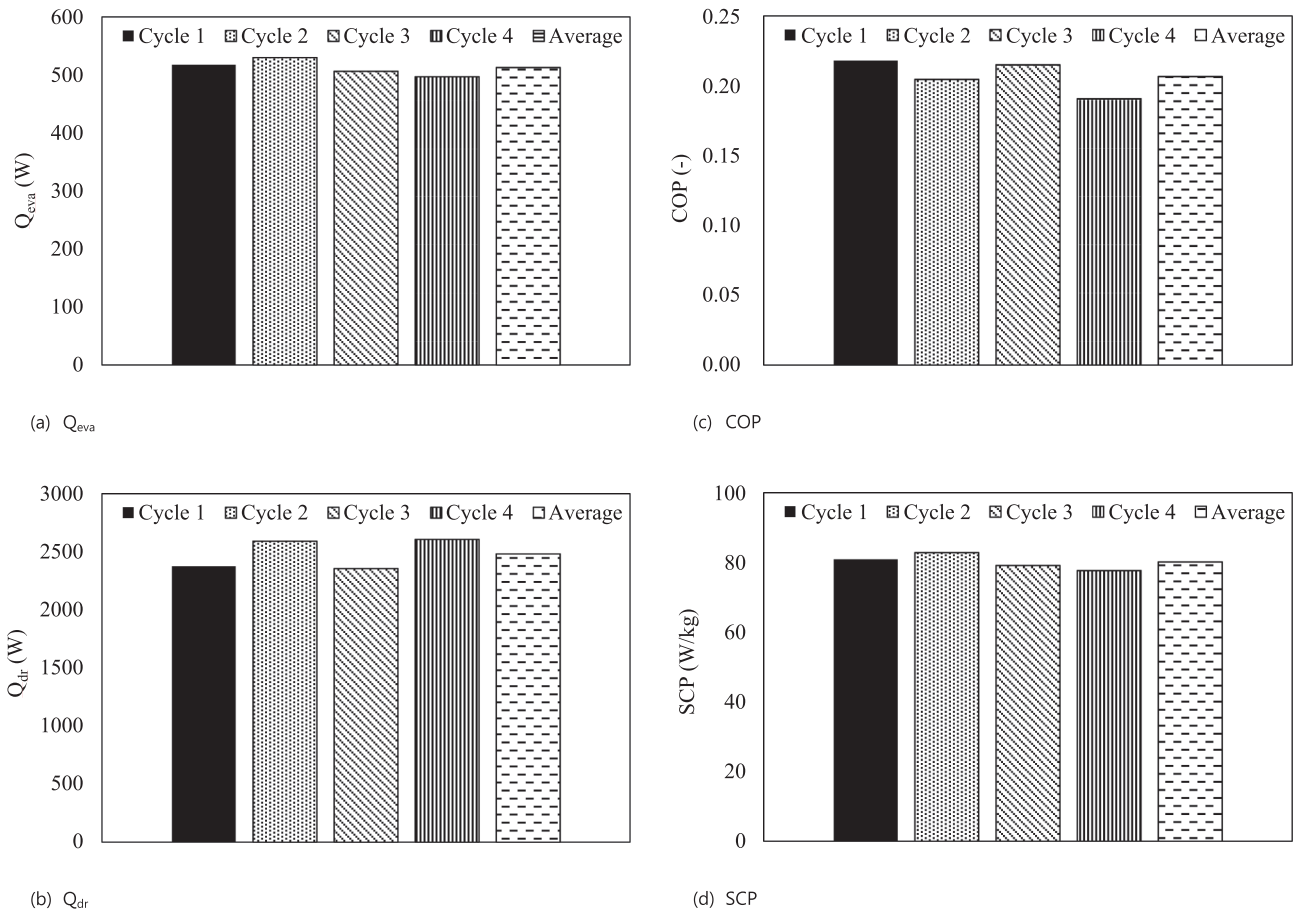


Fig. 6. Average Q_{eva} , Q_{dr} , COP and SCP of each cycle (case 15).

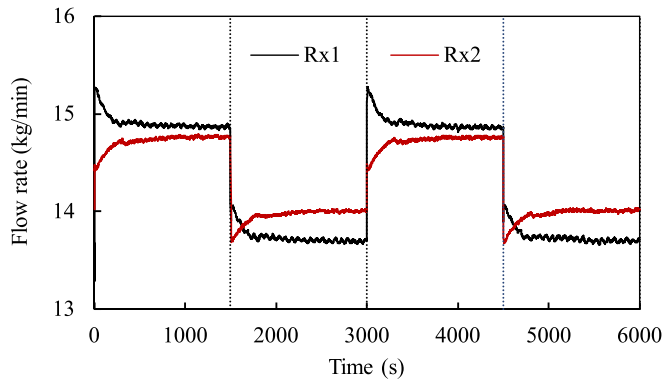


Fig. 7. Flow rates of the secondary fluids in each reactor.

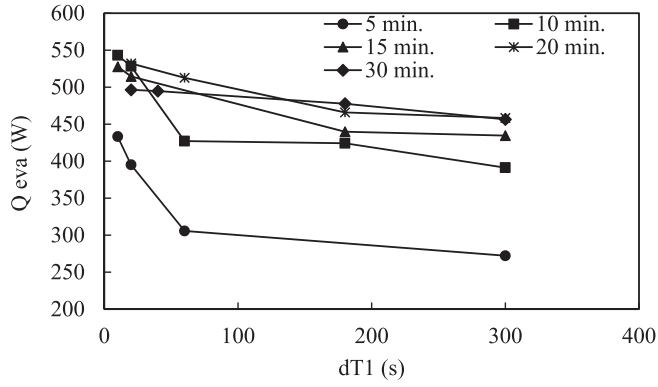
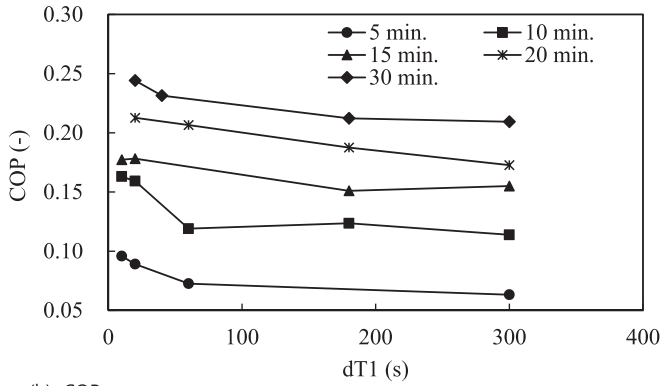
manufactured to be the same, the flow resistance levels in the reactor's header, valve, front and rear pipes have some variation; thus, some deviation of the flow rate occurs even under the same chiller settings. The uncertainties of Q_{eva} , Q_{dr} , COP and SCP for each case were also calculated, and these values for Case 15 are 9.1 W, 47.3 W, 0.005, 1.42 W/kg, respectively. For these reasons, although there are variations in the experimental results depending on the cycle, it can be confirmed that these values are small.

3.3. Performance characteristics of the chemisorption heat pump

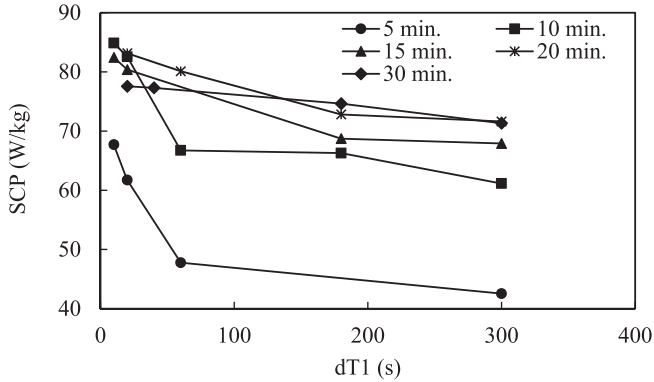
Experiments were conducted under the 21 experimental conditions shown in Table 5. The average Q_{eva} , COP, and SCP values were derived

as described in Chapter 3. Fig. 8 shows the average Q_{eva} , COP, and SCP values with respect to the operation time of the 'Preparation I' mode (dt_1). As shown in Table 5, the total operation time of the 'Preparation I' and 'Preparation II' modes is 300 s. Thus, the operation time of the 'Preparation II' mode in Fig. 8 is 300 s minus dt_1 in each case. There are five legends in Fig. 8, with each referring to the operation time of the 'Adsorption/desorption' mode.

As shown in Fig. 8(a), for all operation times in the 'Adsorption/desorption' mode, Q_{eva} increases as the operation time of the 'Preparation I' mode decreases. The main purpose of the 'Preparation I' mode is to transfer the liquid working fluid from the condenser to the evaporator. When the 'Preparation I' mode starts, the condenser pressure decreases, the evaporator pressure increases, and the liquid working fluid moves from the condenser to the evaporator. At this time, when the pressure of the evaporator rises, the temperature of the fluid inside the evaporator also goes up. Therefore, as the 'Preparation I' mode continues, the heat transfer rate decreases due to a reduction in the temperature difference between the working fluid and the secondary fluid in the evaporator. This explains why the average cooling capacity of the evaporator decreases as the 'Preparation I' mode continues. When the operation time (dt_0) of the 'Adsorption/desorption' mode is five minutes, Q_{eva} increases by about 59 % from 272.2 W to 433.3 W as dt_1 decreases from 300 to 10 s. When dt_0 is 10 min and 15 min, Q_{eva} is improved by 39 % and 21 %, respectively, as dt_1 decreases from 300 to 10 s. When dt_0 is 20 min and 30 min, Q_{eva} increases by 16 % and 9 %, respectively, as dt_1 decreases from 300 to 20 s. The maximum Q_{eva} is approximately 543.2 W and the minimum Q_{eva} is about 272.2 W. It can be seen that the variation of the cooling capacity nearly doubles depending on dt_0 and dt_1 . If the dt_1 is too low, the liquid working fluid in the condenser can accumulate without sufficiently moving to the

(a) Q_{eva} 

(b) COP



(c) SCP

Fig. 8. Average Q_{eva} , COP, and SCP with respect to the operation time of the 'Preparation I' mode.

evaporator. If this phenomenon persists, the liquid working fluid is insufficient in the evaporator, meaning that the continuous alternating operation may not be possible. Therefore, under the condition in which the evaporator or condenser level is kept constant during the alternating operation, the alternating operation should be performed by setting the lowest $dT1$ value possible so that continuous alternating operation and high performance can be obtained. There are optimal operation times of the 'Adsorption/desorption' mode for each $dT1$ condition. When $dT0$ is too low, the cooling capacity is low because an absolutely small amount of refrigerant is adsorbed to the adsorbent. However, when $dT0$ exceeds a certain level, the absolute amount of refrigerant adsorbed on the adsorbent increases, but the adsorption speed decreases. Generally, the adsorption rate is calculated using the Eq. (8) as follow;

$$\frac{dx}{dt} = A \exp\left(-\frac{E_a}{RT}\right) (1-x)^M (\Delta P_{eq})^N \quad (8)$$

As can be seen from the above equation, as the adsorption amount increases, the adsorption rate decreases. For this reason, even if the adsorption amount goes up by increasing $dT0$, the adsorption rate decreases and the average cooling capacity decreases. Therefore, the average cooling capacity tends to increase and then decrease as $dT0$ increases. The optimal time is between 10 and 20 min.

Fig. 8-(b) shows the COP value for each case. As shown in Fig. 8-(b), COP increases upon a decrement of $dT1$ or an increment of $dT0$. The COP of the chemisorption heat pump is defined as Eq. (5). Eq. (5) can be expressed as Eq. (9), where the numerator and denominator refer to the total amount of heat energy absorbed by the evaporator and reactor, respectively.

$$COP_{avg} = \frac{\sum_{i=1}^n Q_{eva,i} \Delta t_{sampling}}{\sum_{i=1}^n Q_{dr,i} \Delta t_{sampling}} \quad (9)$$

The thermal energy supplied to the desorption reactor is used to raise the temperature of the desorption reactor's structure during the 'Preparation I and II' mode and is used as an energy source for the desorption reaction in the 'Adsorption/desorption' mode. Among these, the amount of thermal energy supplied in 'Preparation I and II' mode is almost constant throughout the alternating operation process, regardless of the operation time. On the other hand, the total amount of thermal energy supplied during the desorption reaction process increases as the time for the 'Adsorption/desorption' mode increases. Total amount of thermal energy absorbed by the evaporator is also the same. In other words, if $dT0$ decreases, the proportion of thermal energy supplied to raise the temperature of the structure during the alternating operation process increases, and the proportion of thermal energy supplied for reaction in the 'Adsorption/desorption' mode decreases. Additionally, the amount of heat energy supplied to the evaporator decreases, reducing COP. Thus, COP is high when $dT0$ is longer, as shown in Fig. 8-(b). The maximum and minimum COPs are 0.24 and 0.06, respectively, showing a difference of approximately fourfold.

Fig. 8-(c) shows the SCP value for each case. The SCP is calculated using Eq. (6), meaning that the trends of the results are identical to those in Fig. 8-(a). The minimum and maximum values are 42.5 W/kg and 84.9 W/kg, respectively.

Oliveira and Generoso [24] measured the performance of a chemisorption heat pump system that utilized ammonia and NaBr/EG using a high-temperature heat source of 65–70 °C, a medium-temperature heat source of 28–30 °C, and a low-temperature heat source of 12–15 °C. They suggested that the COP and SCP value of their chemisorption system were 0.32–0.36 and 46.6–64.1 W/kg, respectively. Compared to their research results, the maximum COP and SCP in this study are about 67 % and 132 %, respectively. The performance of the chemisorption heat pump system is affected by the temperature of the high-, medium-, and low-temperature heat sources supplied to the system. Compared to the experimental conditions of Oliveira and Generoso [19], the temperature of the high-temperature heat source under the experimental conditions here is approximately 30 °C lower; that of the medium-temperature heat source is about 10 °C lower, while the results for the low-temperature heat source are similar to each other. The temperature set as the high-temperature heat source in this study results in a relatively unfavorable condition with regard to the performance capabilities, whereas the temperature of the medium-temperature heat source is a favorable condition for performance. Therefore, a direct comparison is difficult, but it can be seen that a similar level of performance could be confirmed when compared with research results in the existing literature [24].

As mentioned earlier, as $dT0$ increases, cooling capacity tends to increase and then decrease. Therefore, depending on $dT0$ and $dT1$, there is an operating point that can operate the system at maximum cooling

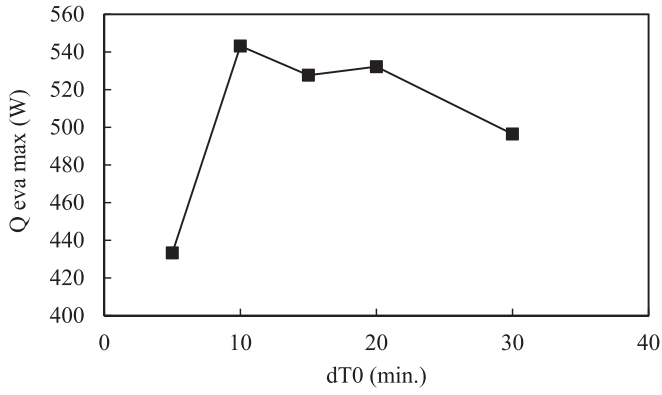
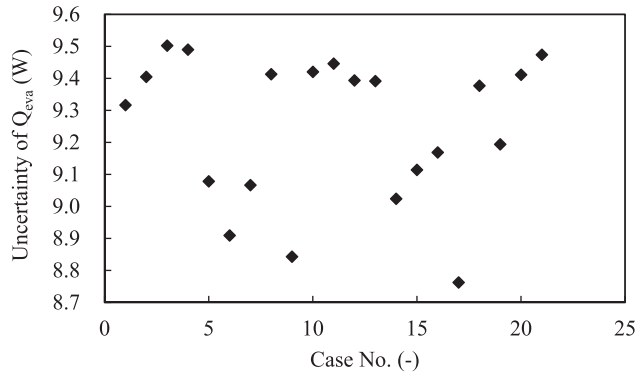
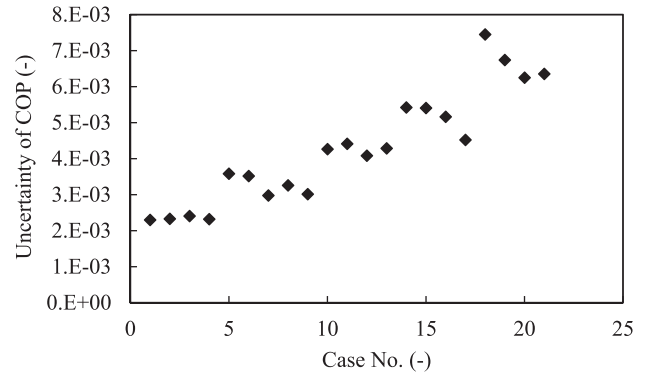


Fig. 9. Maximum average cooling capacity under each operation time of the 'Adsorption/desorption' mode.

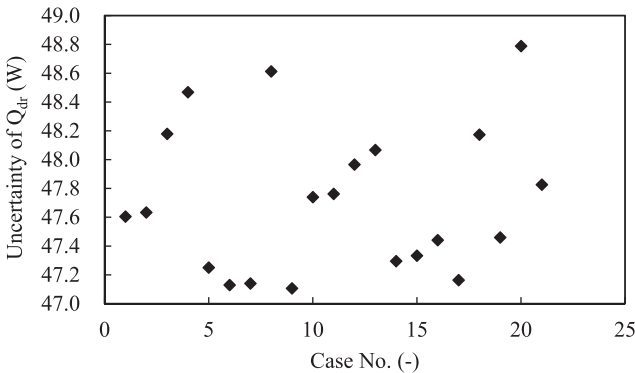
capacity condition. Fig. 9 shows the maximum average cooling capacity under each operation time in the 'Adsorption/desorption' mode in Fig. 8-(a). As shown in Fig. 9, the maximum cooling capacity is 543.2 W when the operation time of the 'Adsorption/desorption' mode is ten minutes and the operation time of the 'Preparation I' mode is ten seconds. Looking at Fig. 8-(b), under the maximum cooling capacity condition, when dT_0 is 10 min, 15 min, and 20 min, COP is 0.16, 0.18, and 0.21, respectively. As can be seen in Fig. 9, the cooling capacity when dT_0 is 10 and 20 min is 543.2 and 532.2 W, respectively, which is only a 2 % difference. However, when dT_0 is increased from 10 to 20 min, COP can be improved by about 30 %, so considering COP, a dT_0 of 20 min can be considered optimal operation point. Although the operating point for optimal operation may be different for each system, higher cooling capacity or higher COP can be achieved by deriving the optimal operating point in this way.



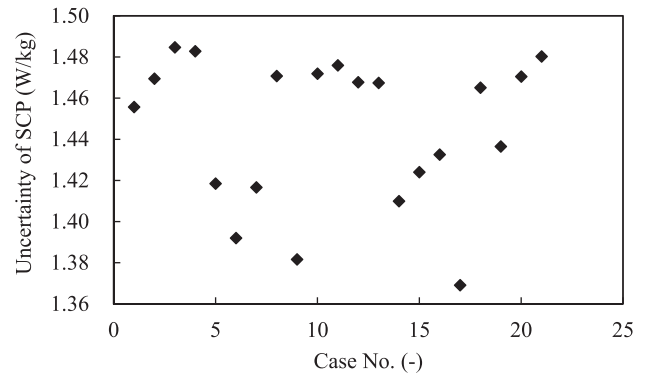
(a) Q_{eva}



(c) COP



(b) Q_{dr}



(d) SCP

Fig. 10. Uncertainties of Q_{eva} , Q_{dr} , COP and SCP for all experimental cases.

Fig. 10 shows the uncertainties of Q_{eva} , Q_{dr} , COP and SCP for all experimental cases. The uncertainties of Q_{eva} , Q_{dr} , COP and SCP are in the ranges of 8.8–9.5 W, 47.1–48.8 W, 0.0022–0.0074 and 1.37–1.48 W/kg, respectively. The average uncertainties of Q_{eva} , Q_{dr} , COP and SCP were 9.25 W, 47.7 W, 0.0043 and 1.44 W/kg, respectively. The uncertainties of Q_{eva} , Q_{dr} , COP can be expressed as a ratio of the measured values in Fig. 8, the values are 1.67–3.5 %, 1.05–2.37 %, and 2.2–3.66 %, respectively. In other words, it can be confirmed that all experimental results were measured with an uncertainty of less than 4 %.

4. Conclusion

In this research, experimental study of a chemisorption heat pump system is conducted. For a chemisorption heat pump system driven by thermal energy at a temperature as low as 40 °C, which has not been considered much in existing literature, the change in system performance according to the operation time of each operation mode was quantitatively analyzed. Chemisorption heat pump with a cooling capacity of 500 W was developed and alternating operations were performed to measure the performance. In order to perform the alternating operation, the flow directions of the working fluid and the secondary fluid were adjusted using an automatic valve for a total of six operation modes. The experiments were conducted for various operation times of the 'Adsorption/desorption' mode and 'Preparation I' mode.

The cooling capacities were varied from 272.2 W to 543.2 under different operation times of the 'Adsorption/desorption' mode and 'Preparation I' mode. The cooling capacity increases as the operation time of 'Preparation I' mode decreases. There is an optimal point at which the cooling capacity is maximized according to the operating time of the 'Adsorption/desorption' mode. When the 'Adsorption/desorption' mode operation time was 10 min and the 'Preparation I' mode operation time was 10 s, the system operated at maximum cooling capacity. At this

time, the cooling capacity was 543.2 W. COP increases as the operation time of the 'Preparation I' mode decreases or that of the 'Adsorption/desorption' mode increases. COP was measured in the range of 0.06–0.24 depending on operating time of each mode. SCPs were measured from 42.5 W to 84.9 kg/W under different operation times in the 'Adsorption/desorption' mode and 'Preparation I' mode.

The operating points at which cooling capacity and COP reach their maximum are different, but considering both variables, the optimal operation point is when the operating time of 'Adsorption/desorption' mode is 20 min and that of 'Preparation I' mode is 20 s. Under this condition, it was confirmed that the COP could be operated at a level of about 0.21 while maintaining 98 % of the maximum cooling capacity.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgments

This work was supported by Korea Institute of Energy Technology Evaluation and Planning (KETEP) grant funded by the Korea government (MOTIE) (20212050100010, Chemisorption heat pump system using electrochemical compressor).

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